Big Data Analytics of Crime Patterns

using Hadoop MapReduce and Apache Hive

Your Group Members

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System Architecture

Pipeline Overview:

1. **Ingestion:** Dataset uploaded to HDFS (`/user/$USER/bigdata_project/crimes`)
2. **MapReduce:** Parallel processing of crime records to aggregate frequencies.
3. **Hive:** Data loaded into Hive via `OpenCSVSerde` for complex SQL analytics.

[Insert Futuristic Architecture Diagram Here]

MapReduce: Crime Type Counting

Goal: Calculate frequency of each primary crime type.

Mapper Logic:

- Parses each CSV line, handling quotes safely.
- Emits `(PrimaryType, 1)`.

Reducer Logic:

- Iterates over values for each key.
- Emits `(PrimaryType, TotalCount)`.

Apache Hive Analytics

Successfully implemented all required SQL clauses for deep insights:

- **SELECT & WHERE:** Filtering narcotic arrests.
- **ORDER BY:** Chronological listing of homicides.
- **GROUP BY & COUNT:** Crimes per location description.
- **HAVING:** Filtering crime types $>$ 500 occurrences.
- **SUM & AVG:** Aggregating total arrests and average domestic crimes.
- **MAX & MIN:** Finding safest and most dangerous districts.
- **JOIN:** Displaying actual community names instead of just codes.

CONCLUSION

Scalable analysis achieved.

THANK YOU